

Resource Management in IoT



Dr. Abhishek Kumar

Problem 1: TDMA Bandwidth & Time Allocation

Q1) Three sensors share TDMA uplink ($T_{\text{frame}} = 100 \text{ ms}$, $B = 1 \text{ MHz}$). Payloads: $S_1=100 \text{ kbits}$ (10 dB), $S_2=200 \text{ kbits}$ (5 dB), $S_3=150 \text{ kbits}$ (20 dB). Is frame feasible? If not:

- (a) Find minimum bandwidth.
- (b) Find maximum payloads in 100 ms (at 1 MHz).

SNR Conversion (dB $\leftrightarrow$ Linear)

$$SNR_{lin} = 10^{\frac{SNR_{dB}}{10}}$$

Shannon rate: $R = B \log_2(1 + SNR)$.

10 dB $\rightarrow$ 10, 5 dB $\rightarrow$ 3.1623, and 20 dB $\rightarrow$ 100

Rates (B=1 MHz):

R1 = 3.459 Mbps, R2 = 2.057 Mbps, R3 = 6.658 Mbps

Required times:

$$t_1 = 28.91 \text{ ms}, t_2 = 97.21 \text{ ms}, \text{ and } t_3 = 22.53 \text{ ms}$$

$$\text{Total} = 148.65 \text{ ms} > 100 \text{ ms} \rightarrow \text{Infeasible}$$

Min bandwidth to fit 100 ms: Since $t_i \propto 1/B_i$,

$$(a) B_{\min} = 1 \text{ MHz} \times (148.65 / 100) = 1.486 \text{ MHz}$$

Max payloads at 1 MHz in 100 ms keeping payload ratios:
Scale by $\alpha = 100/148.65 = 0.67274$

(b) Max payloads (scaled by 0.673):

$$D1' = 67.27 \text{ kbits}, D2' = 134.55 \text{ kbits}, D3' = 100.91 \text{ kbits}$$

Problem 2: Duty Cycling & Lifetime

Q2) A battery-powered IoT mote (3 V) uses:

Active Current: 20 mA for 0.5s per minute (sensing +100ms Tx), Sleep Current: 50 μ A otherwise.

Battery = 2000 mAh **estimate the device lifetime**

Average current:

$$\bar{I} = \frac{I_a \cdot 0.5 + I_s \cdot 59.5}{60} = \frac{20 \text{ mA} \cdot 0.5 + 0.05 \text{ mA} \cdot 59.5}{60} = \boxed{0.21625 \text{ mA}}.$$

Lifetime = $2000 / 0.216 \approx 9248 \text{ h} \approx 385 \text{ days} (\approx 12.7 \text{ months})$

Problem 3: Offload vs Local

A device must process a job requiring 2×10^8 CPU cycles once per second.

Local: CPU $f = 200$ MHz energy per cycle $e_c = 10^{-10}$ J.

Offload: Upload input $D = 500$ kb to edge. Uplink TDMA slot uses $B=1$ MHz, SNR =10 dB, TX power $P = 100$ mW. Assume edge compute/backhaul delays negligible.

Compare total latency and energy per job.

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Local: $t_{loc} = 2 \times 10^8$ cycles @200 MHz $\rightarrow 1.0$ s, $Energy_{loc} = 2 \times 10^8 \times 10^{-10} = 0.02$ J

Offload:

$$R = 1e6 \log_2(1 + 10) = 3.459 \text{ Mbit/s.}$$

Uplink time $t_{\uparrow} = \frac{500k}{3.459M} = 0.1445$ S

TX energy $E_{\uparrow} = P_{t_{\uparrow}} = 0.1 \cdot 0.1445 = 0.01445$ J

Offloading better in both latency and energy.